Fourier Lemmas on the Three-Torus: Heat Smoothing, the L^2 Strain Identity, and the Curl of the Energy Derivative, Formalised in Lean 4

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Abstract

On $\mathbb{T}^3 = (\mathbb{R}/\mathbb{Z})^3$, the product of three circles, we prove in Lean 4 with Mathlib, with no `sorry` and only the standard axioms: the heat semigroup is the Fourier multiplier $e^{-\nu|n|^2 t}$, obeys the semigroup law and contracts L^2; it maps H^s to H^{s+k} with $\|e^{t\nu\Delta} f\|^2_{H^{s+k}} \le k!\, e^{2\nu t}(2\nu t)^{-k}\|f\|^2_{H^s}$; for divergence-free velocity fields $\|S\|_2^2 = \frac{1}{2}\|\omega\|_2^2$; and the kinetic energy, viewed as a function of vorticity, has derivative the stream function, whose curl is the velocity. For the finite-mode Galerkin truncation we further prove local existence, preservation of the Hermitian and divergence-free structure, an H^s transport bound for $s > 5/2$, a uniform-in-N short-time bound, a tame Grönwall propagation estimate, and continuation under an a priori bound. The full build (14 modules, 250 theorems) uses no `sorry` and only the standard axioms. These are Galerkin (finite-mode) and Fourier statements; nothing here proves existence or regularity of Navier–Stokes solutions, and the Clay Millennium problem remains open.

1 Setting

$\mathbb{T}^3$ carries normalised Haar measure; $e_n(x) = e^{2\pi i n\cdot x}$, $n \in \mathbb{Z}^3$, is a Hilbert basis of $L^2(\mathbb{T}^3)$ (Mathlib's `mFourierBasis`), and $\hat{f}(n) = \int f\,\overline{e_n}$.

Lemma 1 (Parseval, `parseval_norm_sq`)**.** $\|f\|_2^2 = \sum_n |\hat{f}(n)|^2$.

Proof. The coefficient map is a linear isometry $L^2 \to \ell^2(\mathbb{Z}^3)$ and $\|a\|_{\ell^2}^2 = \sum |a_n|^2$. □

Lemma 2 (`norm_multiplier_le`)**.** *If $|m(n)| \le 1$ for all n, then $T_m f = \sum m(n)\hat{f}(n)e_n$ defines $T_m : L^2 \to L^2$ with $\|T_m f\|_2 \le \|f\|_2$.*

Proof. $\sum |m(n)\hat{f}(n)|^2 \le \sum |\hat{f}(n)|^2 < \infty$, so the sequence lies in ℓ^2; pull back by the isometry and apply Parseval. □

Commentary. The earlier manuscript already ran the analysis through Fourier coefficients: "The Fourier multiplier proof of the Sobolev embedding $H^s(\mathbb{T}^3, \mathbb{R}^3) \hookrightarrow C^1(\mathbb{T}^3, \mathbb{R}^3)$ for $s > 5/2$ follows from Parseval's identity and the p-series convergence" (*vFinal*, author's unpublished draft). When this build was planned, the author's point was simply: "the torus is proved to be a Circle (cubic)" (30 October 2026). $\mathbb{T}^3$ is three circles, so Mathlib's multidimensional Fourier series and Parseval apply directly, and the Sobolev scale, missing from Mathlib, is defined here as a weighted coefficient sum.

2 Heat semigroup

Theorem 3. *For $\nu, s, t \ge 0$: $\widehat{e^{t\nu\Delta} f}(n) = e^{-\nu|n|^2 t}\hat{f}(n)$, $e^0 = \mathrm{id}$, $e^{s\nu\Delta}e^{t\nu\Delta} = e^{(s+t)\nu\Delta}$, $\|e^{t\nu\Delta} f\|_2 \le \|f\|_2$.*

Proof. The symbol has modulus ≤ 1; $e^{-\nu|n|^2 s}e^{-\nu|n|^2 t} = e^{-\nu|n|^2(s+t)}$ and operators with equal coefficients are equal. □

Commentary. The local-existence block of the May drafts used exactly this operator: "The heat semigroup $e^{\nu t\Delta}$ is bounded on H^s ($s > 5/2$) with $\|e^{\nu t\Delta}\|_{\mathcal{L}(H^s)} \le 1$" (*May 16 edits*, author's unpublished draft). There it was asserted; here it is the contraction theorem above, for every s, since the symbol has modulus at most one at every mode.

3 Sobolev smoothing

$\|f\|_{H^s}^2 = \sum_n (1 + |n|^2)^s |\hat{f}(n)|^2$.

Lemma 4. *For $k \in \mathbb{N}$, $a > 0$, $x \ge 0$: $(1 + x)^k e^{-ax} \le k!\, e^a a^{-k}$.*

Proof. $y^k/k! \le e^y$ with $y = a(1 + x)$, and $e^{-ax} = e^a e^{-a(1+x)}$. □

Theorem 5 (`heat_smoothing`)**.** *For $\nu, t > 0$, $s \in \mathbb{R}$, $k \in \mathbb{N}$, $f \in H^s$: $e^{t\nu\Delta} f \in H^{s+k}$ and $\|e^{t\nu\Delta} f\|_{H^{s+k}}^2 \le k!\, e^{2\nu t}(2\nu t)^{-k}\|f\|_{H^s}^2$.*

Proof. Mode by mode, $(1+|n|^2)^{s+k}e^{-2\nu t|n|^2}|\hat f|^2 \le k!e^{2\nu t}(2\nu t)^{-k}(1+|n|^2)^s|\hat f|^2$ by the lemma with $a = 2\nu t$; sum by comparison. □

Theorem 6 (`heat_smoothing_real`)**.** *The same holds for every real $k \ge 0$ with $k!$ replaced by $\lceil k\rceil!$.*

Proof. For $y \ge 0$, $y^k \le \lceil k\rceil!\, e^y$: if $y \le 1$ then $y^k \le 1 \le e^y$; if $y \ge 1$ then $y^k \le y^{\lceil k\rceil} \le \lceil k\rceil!\, e^y$. The rest is as before. □

Commentary. The v2.4 master draft names this step as indispensable: "Viscosity is essential for local existence in H^s, the explicit higher-norm Gronwall estimate, and the final parabolic smoothing" (*NS Global Regularity v2.4 Master*, author's unpublished draft, 25 May 2026). The drafts invoked the smoothing without a statement; the theorem above gives it with an explicit constant.

4 Strain and vorticity

For $k \in \mathbb{R}^3$, $\hat u \in \mathbb{C}^3$, $k \cdot \hat u = 0$: $\hat S_{ij} = \pi i(k_i\hat u_j + k_j\hat u_i)$, $\hat\omega = 2\pi i\, k \times \hat u$.

Lemma 7. $\sum_{ij}|\hat S_{ij}|^2 = \frac12|\hat\omega|^2$ *and* $|\hat\omega|^2 = 4\pi^2|k|^2|\hat u|^2$.

Proof. Splitting $\hat u = a+ib$, for each real part $\sum_{ij}(k_ia_j+k_ja_i)^2 = 2(|k|^2|a|^2+(k\cdot a)^2)$ and $|k\times a|^2 = |k|^2|a|^2-(k\cdot a)^2$; both $k\cdot a$ and $k\cdot b$ vanish. □

Theorem 8 (`L2_strain_eq`)**.** *For L^2 fields with these coefficients, $\|S\|_2^2 = \frac12\|\omega\|_2^2 \le \|\omega\|_2^2$.*

Proof. Parseval on each component, exchange the finite and infinite sums, apply the lemma per mode. □

5 The curl of the energy derivative

$\hat\psi = \hat\omega/(4\pi^2|k|^2)$, $H(\hat\omega) = |\hat\omega|^2/(8\pi^2|k|^2)$.

Theorem 9. *For $k \ne 0$, $k\cdot\hat u = 0$: $H(\operatorname{curl}\hat u) = \frac12|\hat u|^2$; $H(\omega+\varepsilon\eta) = H(\omega) + \varepsilon\,\mathrm{Re}\langle\hat\psi,\eta\rangle + \varepsilon^2 H(\eta)$, so $\delta H/\delta\omega = \hat\psi$; and $2\pi i\, k\times\hat\psi = \hat u$, i.e.* $\operatorname{curl}(\delta H/\delta\omega) = u$.

Proof. $k\times(k\times\hat u) = k(k\cdot\hat u) - |k|^2\hat u = -|k|^2\hat u$, so $(2\pi i)^2 k\times(k\times\hat u)/(4\pi^2|k|^2) = \hat u$. The expansion is the polarisation of a real quadratic form. □

Commentary. The original degeneracy argument read: "2. $\delta H/\delta\omega = u =$ Biot–Savart ω. 3. By definition of Π_u we have $\Pi_u(u) = 0$" (*Tether_May31.lean*, author's unpublished draft), and the May drafts wrote "$\delta H/\delta\mu = u$ (Biot–Savart recovery, Coulomb gauge)" (*May 16 edits*). Formalising it exposed the slip: as a functional of vorticity, the derivative of the kinetic energy is the stream function ψ, not u, and the ψ-slot tether failed energy degeneracy for that reason (the failure is shown in Lean on an explicit two-mode shear in the public repository `frohmanian-symplectic-tether-regularity`, commit `a52c6af`). The repair was to put the curl into the tether slot, so that what it needs is $\mathrm{curl}(\delta H/\delta\omega) = u$. In the tether build that identity was a hypothesis, because the derivative there is abstract. In Fourier variables it is the theorem above: the earlier statement becomes true once the curl is put back in.

6 Galerkin truncation (finite-mode results)

On the mode box $|k_i| \le N$ the truncated system is a finite-dimensional ODE. Kernel-checked:

- `sobolevProductEstimate`: for $s > 3/2$, $\|fg\|_{H^s} \lesssim \|f\|_{H^s}\|g\|_{H^s}$ with an explicit constant; `summable_fconv_inner`: the convolution series converges absolutely for $f, g \in H^s$, $s \ge 0$.
- `galerkinNS_local_existence` (Picard–Lindelöf) and `galerkinNS_structure_Icc`: Hermitian, divergence-free data stay so on the existence interval.
- `galerkinTransportBound_of_gt`: for $s > 5/2$, $-\mathrm{Re}\langle u, P_N(u\cdot\nabla u)\rangle_{H^s} \le C\|u\|^3_{H^s}$ uniformly in N; `galerkin_uniform_bound_of_gt`: $\|u_N(t)\|^2_{H^s} \le \|a\|^2_{H^s}+1$ on $[0, \texttt{barrierTime}]$, uniformly in N, for Hermitian divergence-free Galerkin solutions.
- `galerkinTameTransportBound_of_gt` and `galerkin_hs_propagation`: if $\|u_N\|_{H^{s_0}} \le A$ ($s_0 > 5/2$) on $[0,T]$ then $\|u_N(t)\|^2_{H^s} \le \|u_N(0)\|^2_{H^s} e^{CAt}$ for $s \ge 2$.
- `galerkinNS_continuation_of_hsBound`: an H^s a priori bound on $[0,T]$ gives a Galerkin solution on all of $[0,T]$.

7 Build status, conditional inputs and scope

Proved. `lake build` of all 14 modules of `TorusFourierNS` succeeds; the axiom report (`AxiomReport.lean`, listed in `AXIOMS.md`) finds 250 theorems,

none using `sorryAx`, and no axiom beyond `propext`, `Classical.choice`, `Quot.sound`. This is a machine check of the Lean statements, not peer review.

Conditional. Results whose hypotheses are not discharged here: the a priori bounds in `galerkin_hs_propagation` and `galerkinNS_continuation_of_hsBound`. The named propositions `SobolevCommutatorEstimate`, `GalerkinUniformHsBound` (for all times) and `GalerkinCompactnessLimit` are stated but not proved.

Open. The range $3/2 < s \leq 5/2$ of the transport bound; passage from Galerkin solutions to solutions of the Navier–Stokes equations; L^p strain bounds for $p \neq 2$; and global existence or regularity of 3D Navier–Stokes, which this work does not address.

A Build and axiom audit

The formal development was built and its axioms audited as follows. This is a build and axiom audit, not peer review.

- Repository:
  `https://github.com/BenFrohman/torus-fourier-ns-lemmas`
- Verified commit:
  `033c80836fa82e5d2d0eab46a76305c2943eec6b`
- Lean toolchain: `leanprover/lean4:v4.30.0-rc1`
- Mathlib: tag `v4.30.0-rc1`, commit (from `lake-manifest.json`)
  `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/torus-fourier-ns-lemmas/actions/runs/41728865393`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/torus-fourier-ns-lemmas/archive/033c80836fa82e5d2d0eab46a76305c2943eec6b.tar.gz`
  `9877a9a9aa2ee4c7271286039bf6b9f71240144f76c1dc87f4d0ce7998d38ae1`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes

itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 205 named theorems and lemmas listed below (the full `AxiomReport.lean` run covers all 250 theorems, with the same result) depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`TorusFourierNS.coeff_eq_mFourierCoeff`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.coeff_multiplier`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.ext_coeff`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.mem p_mul`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.norm_multiplier_le`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.nsq_nonneg`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.parseval_norm_sq`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.summable_coeff_sq`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.two_toReal`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.one_le_sobW`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.summable_fconv_inner`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.summable_fconv_inner_of_gt`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.summable_norm_fconv_inner`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.summable_sq_of_memHs`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.DivergenceFreeL2MeanZero. div_free`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.DivergenceFreeL2MeanZero. mean_zero`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.DivergenceFreeL2MeanZero. summable`	`[propext, Classical.choice, Quot.sound]`
`TorusFourierNS.abs_pairing_le`	`[propext, Classical.choice, Quot.sound]`

Lean declaration	#print axioms
TorusFourierNS.divergenceFree_iff	[propext, Classical.choice, Quot.sound]
TorusFourierNS.fieldCurl_functionalDerivative_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.fieldCurl_stream_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.fieldEnergy_curl	[propext, Classical.choice, Quot.sound]
TorusFourierNS.fieldEnergy_curl_L2	[propext, Classical.choice, Quot.sound]
TorusFourierNS.fieldEnergy_expand	[propext, Classical.choice, Quot.sound]
TorusFourierNS.fieldFunctionalDerivative_energy	[propext, Classical.choice, Quot.sound]
TorusFourierNS.fieldFunctionalDerivative_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.fieldGateauxRep_unique	[propext, Classical.choice, Quot.sound]
TorusFourierNS.fieldPairing_singleDir	[propext, Classical.choice, Quot.sound]
TorusFourierNS.fieldStream_isGateauxRep	[propext, Classical.choice, Quot.sound]
TorusFourierNS.finEnergy_add_smul	[propext, Classical.choice, Quot.sound]
TorusFourierNS.finEnergy_curl	[propext, Classical.choice, Quot.sound]
TorusFourierNS.finEnergy_singleDir	[propext, Classical.choice, Quot.sound]
TorusFourierNS.kvec_ksq_ne_zero	[propext, Classical.choice, Quot.sound]
TorusFourierNS.modeE_curl	[propext, Classical.choice, Quot.sound]
TorusFourierNS.modeEnergy_nonneg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.streamFunctionRepresentative_T3	[propext, Classical.choice, Quot.sound]
TorusFourierNS.summable_coeffField	[propext, Classical.choice, Quot.sound]
TorusFourierNS.summable_pairing	[propext, Classical.choice, Quot.sound]
TorusFourierNS.clampC_dist_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.clampC_norm_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.clampC_of_norm_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.clampG_dist_le	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
TorusFourierNS.clampG_norm_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.clampG_of_norm_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.clamp_real_abs	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinNS_continuation_of_hsBound	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinNS_continuation_of_normBound	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsBoxConst_nonneg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.norm_le_hsNormField_mul	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinHsEnergyInequality_of_transport	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinNSField_apply	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkin_uniform_bound_of_energy	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsNormField_extendG_restrictG_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsNormSq_extendG	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsNormSq_extendG_hasDerivAt	[propext, Classical.choice, Quot.sound]
TorusFourierNS.ode_barrier	[propext, Classical.choice, Quot.sound]
TorusFourierNS.viscCoeff_nonneg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.dot_add'	[propext, Classical.choice, Quot.sound]
TorusFourierNS.dot_galerkinNSField	[propext, Classical.choice, Quot.sound]
TorusFourierNS.dot_neg'	[propext, Classical.choice, Quot.sound]
TorusFourierNS.dot_self_kvec	[propext, Classical.choice, Quot.sound]
TorusFourierNS.dot_smulC	[propext, Classical.choice, Quot.sound]
TorusFourierNS.dot_smulR	[propext, Classical.choice, Quot.sound]
TorusFourierNS.extendG_add	[propext, Classical.choice, Quot.sound]
TorusFourierNS.extendG_apply	[propext, Classical.choice, Quot.sound]
TorusFourierNS.extendG_smul	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
TorusFourierNS.galerkinNS_divergenceFree	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinNS_local_existence	[propext, Classical.choice, Quot.sound]
TorusFourierNS.gconv_add_left	[propext, Classical.choice, Quot.sound]
TorusFourierNS.gconv_add_right	[propext, Classical.choice, Quot.sound]
TorusFourierNS.gconv_smul_left	[propext, Classical.choice, Quot.sound]
TorusFourierNS.gconv_smul_right	[propext, Classical.choice, Quot.sound]
TorusFourierNS.lerayP_add	[propext, Classical.choice, Quot.sound]
TorusFourierNS.lerayP_divFree	[propext, Classical.choice, Quot.sound]
TorusFourierNS.lerayP_smul	[propext, Classical.choice, Quot.sound]
TorusFourierNS.nsBilinCLM_apply	[propext, Classical.choice, Quot.sound]
TorusFourierNS.divergenceFree_extendG_restrictG	[propext, Classical.choice, Quot.sound]
TorusFourierNS.extendG_reflG	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinField_contDiff	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinNSField_contDiff	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinNSField_reflG	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinNS_divergenceFree_Icc	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinNS_hermitian_Icc	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinNS_structure_Icc	[propext, Classical.choice, Quot.sound]
TorusFourierNS.gconv_reflG	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hermitian_extendG_restrictG	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hermitian_of_reflG	[propext, Classical.choice, Quot.sound]
TorusFourierNS.ksq_kvec_neg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.ksq_neg'	[propext, Classical.choice, Quot.sound]
TorusFourierNS.kvec_neg	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
TorusFourierNS.lerayP_conj	[propext, Classical.choice, Quot.sound]
TorusFourierNS.ode_unique_Icc	[propext, Classical.choice, Quot.sound]
TorusFourierNS.reflG_hasDerivAt	[propext, Classical.choice, Quot.sound]
TorusFourierNS.reflG_of_hermitian	[propext, Classical.choice, Quot.sound]
TorusFourierNS.reflG_reflG	[propext, Classical.choice, Quot.sound]
TorusFourierNS.curl_ modeFunctionalDerivative_eq_u	[propext, Classical.choice, Quot.sound]
TorusFourierNS.modeFunctionalDerivative_ eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.modeFunctionalDerivative_ modeEnergy	[propext, Classical.choice, Quot.sound]
TorusFourierNS.modeGateauxRep_unique	[propext, Classical.choice, Quot.sound]
TorusFourierNS.streamSym_isModeGateauxRep	[propext, Classical.choice, Quot.sound]
TorusFourierNS.HsSq_zero	[propext, Classical.choice, Quot.sound]
TorusFourierNS.coeff_heat	[propext, Classical.choice, Quot.sound]
TorusFourierNS.heat.congr_simp	[propext, Classical.choice, Quot.sound]
TorusFourierNS.heat_add	[propext, Classical.choice, Quot.sound]
TorusFourierNS.heat_smoothing	[propext, Classical.choice, Quot.sound]
TorusFourierNS.heat_smoothing_real	[propext, Classical.choice, Quot.sound]
TorusFourierNS.heat_zero	[propext, Classical.choice, Quot.sound]
TorusFourierNS.inHs_zero	[propext, Classical.choice, Quot.sound]
TorusFourierNS.multiplier.congr_simp	[propext, Classical.choice, Quot.sound]
TorusFourierNS.norm_heatSymbol	[propext, Classical.choice, Quot.sound]
TorusFourierNS.norm_heatSymbol_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.norm_heat_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.one_add_pow_mul_exp_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.one_add_rpow_mul_exp_le	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
TorusFourierNS.rpow_le_factorial_ceil_mul_exp	[propext, Classical.choice, Quot.sound]
TorusFourierNS.DivergenceFree.add'	[propext, Classical.choice, Quot.sound]
TorusFourierNS.DivergenceFree.smul'	[propext, Classical.choice, Quot.sound]
TorusFourierNS.MemHsField.add	[propext, Classical.choice, Quot.sound]
TorusFourierNS.MemHsField.sub	[propext, Classical.choice, Quot.sound]
TorusFourierNS.coeff_le_hsNorm	[propext, Classical.choice, Quot.sound]
TorusFourierNS.divergenceFree_of_hs_limit	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkin_local_existence	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsField_complete	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsLp.congr_simp	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsNormField_add_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsNormField_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsNormField_nonneg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsNormField_sub_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsWeighted_add	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsWeighted_sub	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hs_two_toReal	[propext, Classical.choice, Quot.sound]
TorusFourierNS.memHsField_iff	[propext, Classical.choice, Quot.sound]
TorusFourierNS.norm_hsWeighted_sq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.one_add_sq_pos	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sobW_peetre	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sobW_pos	[propext, Classical.choice, Quot.sound]
TorusFourierNS.tendsto_coeff_of_hs	[propext, Classical.choice, Quot.sound]
TorusFourierNS.comp_le_hsNormSq	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
TorusFourierNS.comp_product_bound	[propext, Classical.choice, Quot.sound]
TorusFourierNS.ennCS	[propext, Classical.choice, Quot.sound]
TorusFourierNS.ennCS_fin	[propext, Classical.choice, Quot.sound]
TorusFourierNS.ennr_add_sq_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.ennr_half_sq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.hsNormSq_ennr	[propext, Classical.choice, Quot.sound]
TorusFourierNS.l1_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sobW_neg_le_prod	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sobolevProductEstimate	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sqrt_sobW_neg_mul	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sqrt_sobW_peetre	[propext, Classical.choice, Quot.sound]
TorusFourierNS.summable_one_add_sq_rpow'	[propext, Classical.choice, Quot.sound]
TorusFourierNS.summable_sobW_neg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.tsum_conv_comm	[propext, Classical.choice, Quot.sound]
TorusFourierNS.wh_sq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.young_l1_l2	[propext, Classical.choice, Quot.sound]
TorusFourierNS.young_l2_l1	[propext, Classical.choice, Quot.sound]
TorusFourierNS.L2_strain_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.L2_strain_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.curl_deltaH_eq_u	[propext, Classical.choice, Quot.sound]
TorusFourierNS.curl_sq_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.curl_stream_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.modeEnergy_curl	[propext, Classical.choice, Quot.sound]
TorusFourierNS.modeEnergy_expand	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
TorusFourierNS.normsq_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.strain_sq_eq_half_curl_sq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.strain_sq_le_curl_sq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinTameEnergyInequality_of_gt	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinTameTransportBound_of_gt	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkin_hs_propagation	[propext, Classical.choice, Quot.sound]
TorusFourierNS.tameConst_nonneg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.aC_nonneg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.abs_dotZ_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.abs_mode_le_nrm	[propext, Classical.choice, Quot.sound]
TorusFourierNS.abs_rpow_sub_rpow_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.br_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.br_nonneg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.br_sq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinHsEnergyInequality_of_gt	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkinTransportBound_of_gt	[propext, Classical.choice, Quot.sound]
TorusFourierNS.galerkin_uniform_bound_of_gt	[propext, Classical.choice, Quot.sound]
TorusFourierNS.kpConst_nonneg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.leray_pairing	[propext, Classical.choice, Quot.sound]
TorusFourierNS.modeBox_neg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.norm_triTerm_sub_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.norm_uconj_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.norm_udot_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.nrm_nonneg	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
TorusFourierNS.nrm_sq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.nsq_add	[propext, Classical.choice, Quot.sound]
TorusFourierNS.nsq_add_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.nsq_neg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.preTerm_sum_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.rpow_sub_rpow_le_of_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sig_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sig_neg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sig_nonneg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sig_sq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sobW_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sq_sum_fin3_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.sum_shift_sq_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.transportConst_nonneg	[propext, Classical.choice, Quot.sound]
TorusFourierNS.transport_pairing_eq	[propext, Classical.choice, Quot.sound]
TorusFourierNS.triTerm_sym_cancel	[propext, Classical.choice, Quot.sound]
TorusFourierNS.weight_commutator_le	[propext, Classical.choice, Quot.sound]
TorusFourierNS.young_box	[propext, Classical.choice, Quot.sound]
TorusFourierNS.young_box'	[propext, Classical.choice, Quot.sound]